

Title: SoluPet — AI Builder Case Study: Three-Tier Data + LLM Platform

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Confidentiality: High-level architecture and practices only; no proprietary data, code, or prompts.

Executive Summary

- Built a three-tier platform that turns fragmented source data into reliable features, powers ML forecasts, and ships two production LLM apps with strict output contracts and HITL governance.
- Stack: Azure + Databricks (PySpark, Python, SQL), GPT-4o, embeddings, JSON schema enforcement.
- Impact: ~40–50% less manual analysis and ~50% faster time-to-insight across weekly executive operations.

Tier 1 — Data Foundation (Azure + Databricks)

Objectives

- Create reproducible, incrementally updated datasets for ML/RAG with lineage and data quality guarantees.

Core Patterns

- Medallion architecture: Bronze (ingest) → Silver (conform) → Gold (features/serving).
- PySpark pipelines with parameterized configs; idempotent, partitioned, and incremental writes.
- Data quality gates: schema drift detection, distribution/volume checks, null/dup constraints.
- Feature engineering for time series: lags, rolling stats, seasonality flags, entity joins.
- Observability: run metadata, data SLAs, and cost/perf tuning (Z-ordering, caching).

Deliverables

- Curated Gold tables for forecasting and retrieval; data lineage and versioned schema registry.

Tier 2 — Model Intelligence (Forecasting + LLM/RAG)

Forecasting Track

- Trained time-series models on Gold features; backtesting with sliding windows and error bands.
- Stored predictions with confidence intervals and model/version metadata for downstream use.

LLM Track (GPT-4o + Embeddings)

App 1 — Metric Synthesis

- Purpose: Condense large metric volumes into executive-ready narratives with anomaly callouts.
- Design: Retrieval over curated metrics; prompt scaffolds with role/constraint blocks; temperature control.

App 2 — Recommendation Extraction

- Purpose: Convert analysis artifacts into structured, actionable recommendations.
- Design: Embedding-based context assembly; strict JSON schemas; toolformer-style instructions for function calling when applicable.

Reliability & Governance

- JSON schema validation and retry-on-invalid; fallbacks to minimal-safe responses.
- Human-in-the-loop (HITL) checkpoints for critical outputs; audit logs of prompts/responses.
- Eval harness: unit prompts, regression sets, and periodic blind reviews to track drift/hallucination.

Tier 3 — Decision Enablement (Serving + Controls)

Delivery

- Scheduled reports and dashboards backed by Gold tables and prediction/LLM outputs.
- API endpoints for programmatic access; role-scoped summaries vs. analyst drill-downs.

Controls

- Versioned prompts and models with change approvals.
- Access aligned to least privilege; PII-aware handling when applicable.
- Runbooks for incident response (schema change, model degradation, LLM invalid output).

Measured Outcomes

- ~40–50% reduction in manual analysis tasks.
- ~50% faster time-to-insight for weekly executive operations.
- Coverage: insights used in 100% of weekly executive reviews.

Architecture Snapshot

- Ingest → Bronze/Silver/Gold → Forecasting + RAG/LLM → Validation (JSON + HITL) → Dashboards/APIs.
- Telemetry across all layers: data quality, model accuracy, LLM validity, latency, and cost.

Operational Practices

- CI/CD for notebooks, prompts, and config; environment promotion (dev → test → prod).
- Prompt/version management: semantic diffing, rollback, and offline eval before deploy.
- Cost controls: batching, token budgeting, caching embeddings/context windows.
- Safety: constrained decoding where possible, Pydantic-style schema enforcement, guardrails for tool use.

Limitations & Next Steps

- Strengthen automated red-teaming and coverage-based evals for LLM updates.
- Expand feature store contracts and add auto-retraining triggers based on drift.
- Broaden recommendation ontology and plug in retrieval evaluators (precision/recall on contexts).

Skills Demonstrated

- Data engineering at scale (Azure, Databricks, PySpark, SQL).
- LLM application engineering (RAG, embeddings, prompt/response contracts, evals).
- MLOps/LLMOps (observability, governance, CI/CD, HITL).
- Product mindset: measurable impact, stakeholder-aligned outputs, reliability in production.

Note: This case study is process-oriented and omits any confidential data or implementation specifics.